You Can Just Do Things

You Can Just Do Things

A Data Portrait of John D. Cook's Blog

Owen D. Goode

2025

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All statistics in this book reflect the blog as of November 27, 2025.

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For John

Foreword

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You Can Just Do Things

5,233 posts.

Roughly one every 1.25 days, from January 2008 through November 2025. No breaks. No sabbaticals. No pivots to video, no podcast spinoffs, no Substack migration. Just a mathematician writing about what interested him, day after day, for nearly two decades.

This is a book about what that looks like.

Why care about a mathematician's blog?

Because it demonstrates something increasingly rare: the ability to just do things. To write about interests, publish without permission, build utility over time, and—without virality, platform, or strategy—become genuinely influential.

John D. Cook's blog appears in university syllabi at MIT, Stanford, and Carnegie Mellon. Its code snippets run in production systems at companies Cook has never heard of. When a programmer searches "gamma function python" at 2 AM, trying to debug something urgent, they find him. When a graduate student needs an intuitive explanation of Bayesian inference, they find him.

The numbers hint at the reach: nearly 6,000 citations on Google Scholar. Over 1,200 submissions to Hacker News, accumulating 25,000 upvotes. Readers in 244 countries. A single interpolation calculator page has generated over 600,000 search clicks in the most recent 16-month Google Search window.

This is influence without fame. Infrastructure without brand. It was built simply by showing up and explaining things, day after day, for seventeen years.

The method applies beyond mathematics. Whatever you know deeply—whatever you could explain to someone who needs to understand—you can build the same kind of quiet influence. You don't need permission. You don't need a platform. You just need to start, and then keep going.

What would the shape of your own work look like, if someone counted and charted it seventeen years from now?

The blog launched in January 2008—the year the iPhone App Store opened,

the year Twitter was still asking "what are you doing?" instead of optimizing for engagement. The internet of 2008 was a different place. People had blogrolls. Google Reader existed.

While the internet reinvented itself three times over, johndcook.com kept doing one thing. Math. Computing. Statistics. The occasional digression into music theory or typography. No ads. No viral moments. No controversy. No brand.

There's a term for this in ecology: a *{*climax species*}*, the organism that remains after the initial colonizers have come and gone, after the boom-and-bust cycles have played out, after the ecosystem has stabilized. The internet's climax species are not the viral sensations. They're the quiet sites that just keep being useful.

This book attempts to see the shape of that persistence.

Not by reading every post—though many were read—but by counting, measuring, and visualizing. How many posts about prime numbers? (103.) What day does he publish most? (Tuesday.) How has the average post length changed? (It doubled.) When did cryptography become a major topic? (2019.)

The data reveals patterns. The patterns suggest stories. The stories, taken together, form a portrait—not of a person, but of a practice. A way of showing up. A way of thinking in public.

This is not a biography. John D. Cook's personal life isn't the subject. This is a data portrait of sustained intellectual work. It asks what it looks like when someone just *{*keeps going*}*.

A few things this book is not:

It's not comprehensive. 5,233 posts cannot be summarized. The chapters that follow trace patterns and highlights, but most posts go unmentioned.

It's not evaluative. The data can show which posts are longest or most tagged—but not which are most valuable. That belongs to readers.

It's not authorized. John D. Cook didn't commission this book. It's a gift, made from public data, offered in appreciation.

What it is: an experiment in seeing. Can a body of work be understood from above? Can numbers and charts reveal something that reading alone would miss?

More than seventeen years. 1.8 million words—the equivalent of eighteen novels, or Shakespeare twice over. One voice, steady and curious, writing into the void and finding that the void was full of people who needed exactly what he had to offer.

This is the shape of that work.

All statistics in this book reflect the blog as of November 27, 2025. The archive continues to grow.

Chapter 1

The Shape of Seventeen Years

The first post appeared on January 9, 2008: Moore's law and software bloat, 182 words. A small observation. Nothing viral.

The most recent post, as of this writing, appeared on November 27, 2025: four equivalent formulas for measuring distance in hyperbolic space, with Python code. Still small observations. Still no announcements.

The numbers are simple: 5,233 posts over 6,531 days. The shape of those numbers is the story.

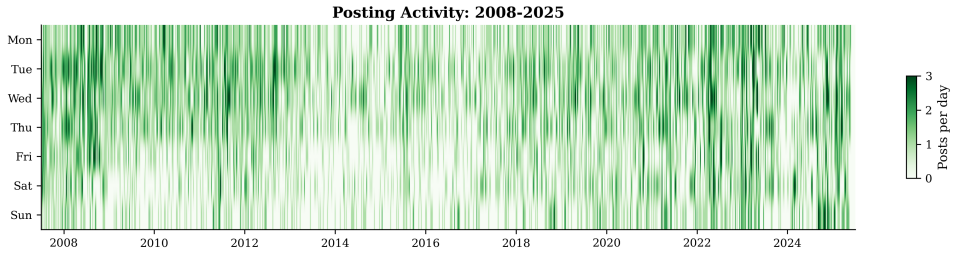


Figure 1.1: Posting activity across 17 years

Year by Year

The shape of the blog changes over time, but the fact of its presence does not. The annual accounting:

Year	Posts	Avg Words	Notes
2008	410	206	Launch year—highest volume
2009	366	256	
2010	284	268	
2011	317	222	
2012	357	216	
2013	260	230	
2014	147	188	Lowest year
2015	168	293	Recovery begins
2016	209	386	Posts getting longer
2017	184	383	
2018	266	473	Length nearly doubled from early years
2019	288	446	
2020	288	451	Pandemic year—no change in rhythm
2021	341	451	
2022	362	476	Peak average length
2023	395	435	
2024	248	411	
2025	343	436	(partial year through November)

The total count obscures the real story.

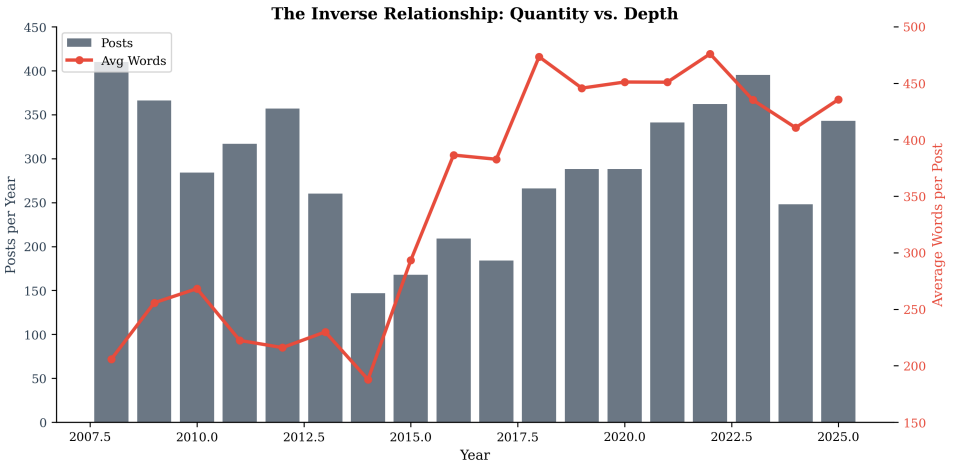


Figure 1.2: The inverse relationship: fewer posts, more depth

The Launch Year

2008 stands alone. Four hundred and ten posts in a single year—more than one per day on average. Cook posted twice on January 9th, his first day. He posted on January 10th. And 11th. And kept going.

This is the energy of beginning. A new blog, a new outlet, ideas that had been accumulating for years finally finding a place to go. The posts are short—206 words on average, some under 100. They read like notes jotted between other work. A quick observation about software bloat. A pointer to an interesting integral. A link to someone else's insight.

Many new blogs start this way. Few sustain it. The typical pattern is a burst of enthusiasm followed by gradual abandonment. The internet is littered with blogs that have three posts from 2009 and then nothing.

Cook's first year was not a burst. It was the beginning of a practice.

The Quiet Year

Then there's 2014. One hundred forty-seven posts—barely one every two and a half days. The average word count dropped to 188, the lowest of any year. Something changed.

The numbers don't say why. The blog offers no announcement, no "I'm taking a break," no explanation of reduced capacity. The posts simply came less frequently for a while, and then more frequently again.

Looking month by month, the reduction is distributed evenly across the year. January: 9 posts. February: 15. March: 11. No single month stands out as empty; no dramatic event visible in the rhythm. Just a year with less to say, or less time to say it, or both.

The surrounding years provide context. 2013: 260 posts. 2014: 147. 2015: 168. The dip begins in 2014, continues into 2015, and then the numbers climb again. By 2016, the blog produces 209 posts—and crucially, those posts are significantly longer. The average jumps from 188 words to 386.

Whatever happened in 2014, the blog emerged from it transformed. The years of many short posts gave way to fewer but deeper investigations. The practice adapted.

The Trade-Off

Here is the central pattern of seventeen years: an inverse relationship between quantity and depth.

The early era (2008-2012) produced 1,734 posts averaging 232 words each. The middle era (2013-2017) produced 968 posts averaging 297 words. The recent era (2018-2025) produced 2,531 posts averaging 448 words.

The recent era produced over 1.1 million words—about 1.6 times as many as the first two eras combined—despite spanning fewer years. The blog became both more prolific and more substantive. More posts *{*and*}* longer

posts. The assumed trade-off—depth versus frequency—turned out not to apply.

This is what sustained practice looks like. The early years were learning the form. The middle years were consolidation. The recent years are mastery—the ability to produce more, with more substance, without apparent strain.

The Gaps

Every blogger takes breaks. The question is how long.

The longest gap in the entire archive is 11 days between posts on July 9 and July 21, 2014. The second-longest gap is 9 days, in May 2014.

No gap reaches two weeks. No sabbatical. No "I'm back after six months." For nearly eighteen years, the blog has never gone silent for more than eleven days.

Most professionals take longer vacations than the blog's longest gap. A two-week holiday would set a new record. A month away would be unprecedented. Whatever else was happening in Cook's life—travel, illness, family, career changes—the blog kept appearing.

The pattern isn't workaholism. The posting frequency is measured, not manic. It's something else: a commitment to showing up, week after week, year after year, regardless of mood or circumstance. The practice survives because it's designed to survive.

The Pandemic Invariance

March 2020 upended most patterns. Schools closed, offices emptied, conferences cancelled. The world's rhythms broke.

The blog's rhythms did not.

2020: 288 posts, averaging 451 words. The monthly distribution is unremarkable—22 posts in March (the month everything changed), 21 in April, 23 in May. No spike of anxiety-posting. No drought of distraction. Just another year, indistinguishable from 2019 and 2021.

This flatness is extraordinary. It suggests that the practice serves as a sanctuary—a place where the chaos of the world is ordered into logic.

The blog rarely comments on the pandemic directly. Aside from a handful of posts that use COVID-19 data as statistical examples, it simply continues—another post about probability, another post about Python, another post about number theory. The world outside the blog changed. The blog's function did not.

What Changed

Compare the first post to a recent one.

January 9, 2008, "Moore's law and software bloat": A 182-word observation that new software bloats to consume new hardware speed. The post links to one external article. It offers no code, no equations, no deep investigation. It's a thought, shared.

November 27, 2025, "Equal things that don't look equal": A discussion of four different formulas for measuring distance in hyperbolic space, with Python code demonstrating their equivalence, mathematical notation rendered in LaTeX, links to previous posts for context, and a follow-up update about numerical precision. The depth is evident.

Same author. Same platform. Same basic form. But the execution has evolved dramatically. The recent posts assume more context, provide more code, connect more explicitly to the rest of the blog's ecosystem. They read less like notes and more like chapters in an ongoing textbook.

What Didn't Change

Still, the core pattern persists. Short, focused posts. One topic per post. A mix of math, computing, statistics, and curiosity. No personal updates, no industry hot takes, no engagement bait. Just: here's something interesting. Here's how it works. Here's some code.

The topics shift over seventeen years—cryptography emerges, PowerShell fades, Python dominates—but the *{*kind} of thinking doesn't change. A working mathematician explaining things to himself, in public, for anyone who happens to be interested.

Cook's own blog provides the frame. In a 2025 post, he observes: "The limiting factor on statistics projects has rarely been my knowledge of statistics. The limiting factors are things like communication, endurance, organization."

Endurance. Organization. The words describe the blog itself.

Reading the Shape

Seventeen years of data trace the outline of a practice.

Not a career, though it intersects with one. Not a brand, though others know the name. Not a product, though the writing has value. A practice: something done regularly, for its own sake, getting better over time.

The shape is not linear. It dips in 2014 and rises after. The shape is not stable. Post length doubles over the middle years. The shape is not heroic. Most posts are short, most days uneventful, most numbers unremarkable.

But the shape is complete. Every year filled in. Every month represented. An activity heatmap with no empty regions, no abandoned stretches, no evidence of giving up.

5,233 posts. 1.8 million words. 6,531 days. One voice, present and accounted for.

What does sustained intellectual output look like? This.

The shape is established. The next question is what fills it—what topics earn that sustained attention, chapter after chapter, year after year.

Chapter 2

The Topography of Interest

> "The blog's twenty-six categories are the closest thing Cook has to a table of contents for his own mind."

The blog has categories. WordPress requires it—you cannot publish a post without assigning it to at least one category. Many bloggers treat this as a chore, clicking the obvious option and moving on.

Cook has twenty-six categories. Taken together, they form a map of his mind.

The Overwhelming Center

Math: 2,378 posts. Forty-five percent of everything.

Not "mathematics," which would sound academic. Not "maths," which would sound British. Just "Math"—the working term, the doing word. And it dominates the blog like nothing else dominates anything.

To visualize the proportions: if the blog were a country, Math would be the territory west of the Mississippi. Everything else would share the East.

The second-largest category is "Uncategorized" at 996 posts (19%). We'll return to this mystery. But the named categories reveal a steep power law: Math dominates everything, while the practical categories—Computing, Software Development, Statistics—form a second tier that serves the first.

Here is the breakdown:

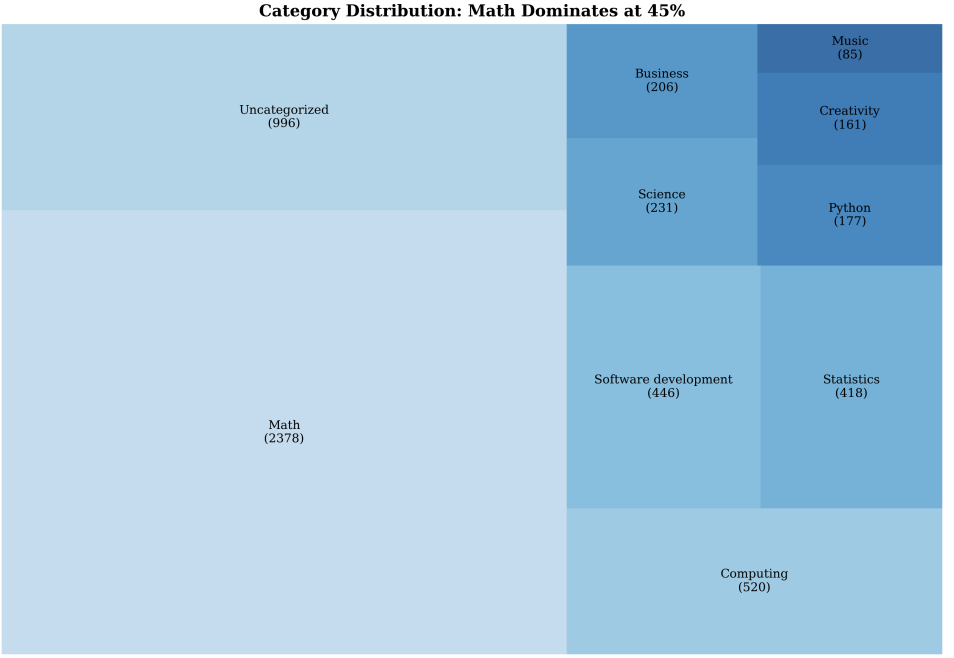


Figure 2.1: Category distribution: Math dominates at 45%

Category	Posts	% of Total
Math	2,378	45.4%
Uncategorized	996	19.0%
Computing	520	9.9%
Software development	446	8.5%
Statistics	418	8.0%
Science	231	4.4%
Business	206	3.9%
Python	177	3.4%
Creativity	161	3.1%
Music	85	1.6%
Clinical trials	55	1.1%
Typography	50	1.0%
PowerShell	47	0.9%
Graphics	43	0.8%
Geography	30	0.6%
Machine learning	24	0.5%
AI	16	0.3%
Algorithms	15	0.3%
(8 more minor categories)	…	<0.2% each

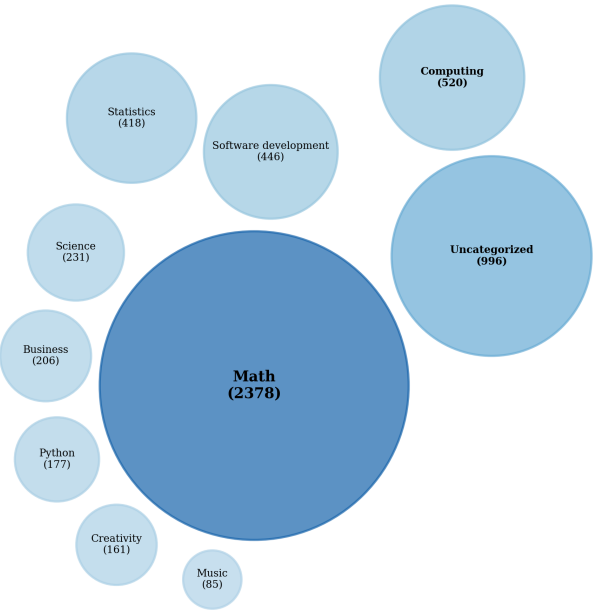


Figure 2.2: The category universe

What Kind of Math?

"Math" is a big tent. The posts within it span everything from recreational puzzles to graduate-level analysis. The tags within Math posts reveal the internal structure:

The top ten tags in Math posts: 1. Number theory (267 posts) 2. Probability and Statistics (250) 3. Special functions (134) 4. Differential equations (124) 5. Geometry (103) 6. Complex analysis (90) 7. Cryptography (79) 8. Numerical analysis (64) 9. Integration (57) 10. Linear algebra (55)

Number theory leads—the study of integers, primes, divisibility, the atoms of arithmetic. This is the purest of pure mathematics, the branch G. H. Hardy famously defended as having "no practical applications." That it leads the tags suggests Cook's heart lies with fundamentals.

But look at the mix that follows: probability, differential equations, numerical analysis. It is not the pure mathematician's curriculum; it's the applied mathematician's toolkit. The blog sits at the intersection of rigor and utility.

The Practical Layer

Below Math in the hierarchy sits a cluster of practical categories:

- **{**Computing} (520 posts, 9.9%) - **{**Software development} (446 posts, 8.5%) - **{**Statistics} (418 posts, 8.0%)

Together these constitute more than a quarter of the blog. They overlap with Math frequently—91 posts carry both Computing and Math categories, 56 share Math and Statistics—but they signal a distinct audience: people who write code, build systems, analyze data.

Notice what's absent from this practical layer: trendy buzzwords. "Machine learning" has only 24 posts. "AI" has only 16. The blog predates the 2012 deep learning revolution, and while it occasionally touches on neural networks and language models, it doesn't chase the hype cycle. The practical layer is timeless practical—shell scripting and probability distributions, not prompt engineering.

The Surprising Corners

The small categories reveal more than the large ones.

{Music} (85 posts, 1.6%): A mathematician writing about music is not unusual. But 85 posts is significant. These aren't casual mentions; they're sustained investigations. "Circle of fifths and number theory." "The acoustics of kettledrums." "How many musical scales are there?" Nineteen of the Music posts also carry the Math category. When Cook writes about music, he's doing mathematics.

{Typography} (50 posts, 1.0%): An unexpected obsession. Posts about accented letters in HTML, typesetting music notation, LaTeX commands for obscure symbols, the rendering of double-struck capital letters. These are not mathematical topics. But they reveal something about the author: a care for how symbols appear on the page, a precision that extends beyond content to presentation.

{Clinical trials} (55 posts, 1.1%): This category has a strange shape. The posts span 2008-2018, then stop. Completely. Not a gradual fade—a hard cut. Something happened in 2018. The data doesn't say what. But the pattern is visible: ten years of writing about clinical trial design, Bayesian adaptive methods, cohort assignments—and then nothing. A closed chapter.

{Geography} (30 posts, 0.6%): Small but persistent. "Zip codes, geocodes, and Hilbert curves." "State abbreviations as a random walk." The spatial imagination of a mathematician applied to maps and territories.

The Uncategorized Mystery

996 posts escape the author's own taxonomy. Nearly one in five.

What are they? A sample reveals the pattern: they're not any one thing. They're everything else.

- "Productivity advice from Mark Twain" — a quote, filed nowhere - "AI Spring" — a brief observation without sustained analysis - "Breach Safe Harbor" — privacy commentary that doesn't fit Computing - "Example of memorizing a 256-bit private key" — cryptography crossed with memory techniques - "Impulse response" — differential equations, but not quite Math

The Uncategorized category is a miscellany bin. It holds 470 posts with no tags at all—thoughts that arrived without classification. It holds cross-disciplinary observations that span boundaries. It holds brief pointers and links.

Consider "The acoustics of your desk" (2019), filed under Uncategorized. The post explains how a desk's resonant frequency affects what you hear when you knock on it—physics, yes, but also acoustics, also woodworking, also the mathematics of vibration. Where does it belong? Nowhere exactly. Everywhere loosely. So: Uncategorized.

The category's size suggests something deeper: the limits of any taxonomy. Twenty-six categories cannot capture the full range of a working mind. When the thoughts don't fit, Cook publishes them anyway. The "Uncategorized" posts are evidence of intellectual activity that resists sorting—the overflow of curiosity.

The Overlaps

621 posts carry multiple categories. These intersections map the author's mental geography.

The top five combinations: 1. Computing + Math (98 posts) 2. Math + Statistics (64) 3. Math + Python (58) 4. Math + Science (37) 5. Math + Software development (34)

Math is the hub. Almost every significant combination includes it. The blog's network structure is a star topology with Math at the center.

But some combinations are unexpected:

- **{**Math + Music} (19 posts): Where scales become ratios and harmonics become Fourier series. "The opening chord of 'A Hard Day's Night'" analyzed mathematically.

- **{**Clinical trials + Statistics} (22 posts): The professional overlap, where Bayesian methods meet drug development.

- **{**Business + Software development} (26 posts): Where technical skill meets organizational reality. Consulting, project management, the human side of technical work.

- **{**Geography + Math} (15 posts): Where maps become mathematical objects. Coordinate systems, projection problems, geocoding algorithms.

The overlaps reveal what the single categories miss: a mind that doesn't respect disciplinary boundaries. The blog is not a math blog that occasionally mentions code, nor a coding blog that occasionally invokes math. It lives in the intersection permanently.

Evolution of the Territory

The categories are not static. Their proportions shift over the seventeen years.

Math has grown both in absolute and relative terms: - 2008: 85 posts (21% of that year's output) - 2012: 73 posts (20%) - 2016: 104 posts (50%) - 2020: 199 posts (69%) - 2024: 138 posts (56%)

The blog has become more mathematical over time. What began as a balanced mix tilted toward Math's dominance, especially after 2016.

Meanwhile, Statistics has declined: - 2008: 81 posts - 2012: 33 posts - 2020: 10 posts - 2024: 16 posts

This may reflect career changes. The clinical trials category died in 2018; Statistics, its close companion, faded with it. The professional statistician's work shifted, and the blog's categories followed.

Python as a category shows another pattern: - 2008: 10 posts - 2012: 23 posts - 2016: 10 posts - 2020: 5 posts

The Python category peaked in the early 2010s and declined—not because Python posts stopped, but because Python became so embedded in Math and Computing that separate categorization seemed redundant. The tag "Python" appears in 83 Math posts; the language has been absorbed into the mathematical practice itself.

Reading the Categories

A taxonomy is a theory of what matters. When Cook created twenty-six categories, he made twenty-six claims about the structure of his own interests.

The resulting map has features:

{A single dominant territory.} Math is not first among equals; it's an empire. Everything else exists in relation to it.

{A practical belt around the center.} Computing, Software development, Statistics—these surround Math and connect it to the working world. They're not peripheral; they're infrastructure.

{Small holdings at the edges.} Music, Typography, Geography—tiny territories that persist year after year. Not big enough to shape the whole, but present enough to signal their importance.

{A large unclaimed wilderness.} Nearly a fifth of the blog escapes categorization. The taxonomy is incomplete by design, or by necessity.

This is what a mathematician's mind looks like when laid out in categories: one overwhelming center, a practical ring, scattered outposts of curiosity, and a significant portion that refuses to be mapped.

Beneath the Map

The category data offers an answer to what a mathematician's mind organizes itself around: mathematics, but not only mathematics. Around the intersection of rigor and application. Around sustained curiosities that persist for decades. Around the willingness to write about whatever proves interesting, whether or not it fits.

The topography is not neat. It's not the curriculum of an academic department. It is one person's working map of what matters, updated over seventeen years, with all the irregularity that implies.

Cook once quoted a passage about category theory in mathematics: it "clears away tangled multitudes of individually trivial problems" and "puts the hard problems in clear relief." The blog's own categories do something similar.

It's about math. It's about making math work. It's about everything that touches math when you're not being too careful about boundaries.

The twenty-sixth category—Uncategorized—is the most honest one. The mind that wrote this blog does not fully fit its own taxonomy.

Chapter 3

The Rhythm of a Working Mind

> "I've never had a schedule for this blog: I write when I feel like writing, which usually means several times a week."
>
> "Blogging pace" (2024)

Temporal patterns are autobiography written in timestamps. Every post carries a date and time. Aggregate 5,233 of them and a portrait appears of when one person does his thinking.

The Weekly Shape

Tuesday is the peak.

Day	Posts	%
Tuesday	1,003	19.2%
Wednesday	938	17.9%
Monday	887	17.0%
Thursday	823	15.7%
Friday	561	10.7%
Saturday	555	10.6%
Sunday	466	8.9%

The week has a clear arc. Activity rises from Monday through Tuesday, holds through Wednesday, then falls through the weekend. Friday sees a notable drop—down to 10.7%, barely more than Saturday. By Sunday, posting reaches its minimum.

The weekday-to-weekend ratio is 4.13:1. For every post that appears on Saturday or Sunday, more than four appear Monday through Friday.

The shape isn't a hobbyist's pattern. A hobbyist—someone who blogs in their spare time, after work and on weekends—would show the opposite shape: peaks on Saturday and Sunday, valleys during the workweek. The

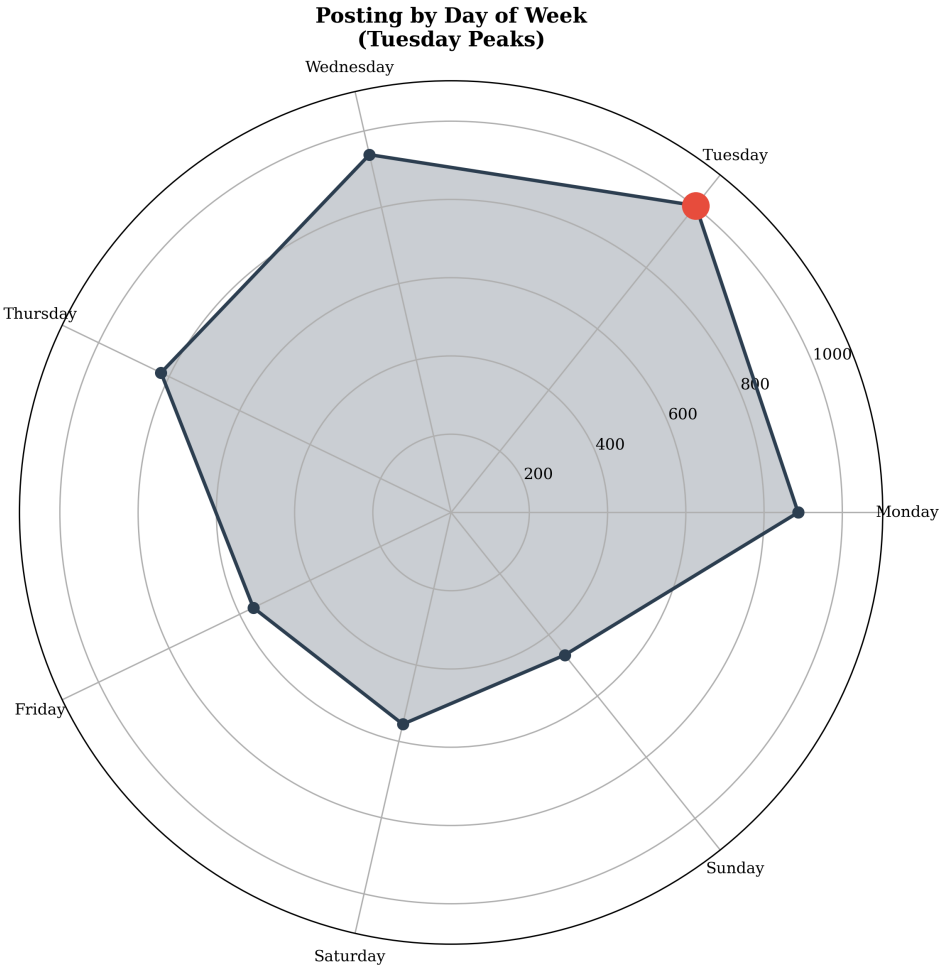


Figure 3.1: Posting by day of week: Tuesday peaks

data shows a professional's schedule: blogging is part of the work, and work happens on workdays.

But why Tuesday? Why not Monday, the fresh-start day? Why not Friday, when the week's thoughts have accumulated?

The data alone doesn't explain it. But a plausible story: Monday is for meetings, email, catching up on what accumulated over the weekend. Tuesday is the first day clear enough for sustained thought. The blog posts that appear on Tuesday may be written on Tuesday, or they may be the refined versions of thoughts that began on Monday and crystallized overnight. Either way, Tuesday is when the typing happens.

The Daily Shape

The hourly distribution tells a clearer story:

Hour	Posts	%
7:00 AM	683	13.1%
8:00 AM	580	11.1%
6:00 AM	501	9.6%
9:00 AM	503	9.6%
10:00 AM	323	6.2%
11:00 AM	315	6.0%
…	…	…
3:00 PM	194	3.7%
6:00 PM	139	2.7%
11:00 PM	19	0.4%

This is a morning person.

Nearly 44% of all posts appear between 6:00 AM and 10:00 AM. The peak hour is 7:00 AM, with 683 posts—more than one in eight. Add the 6:00 AM hour and more than one in five posts appears before most people have finished breakfast.

After 10:00 AM, the numbers decline steadily through the afternoon. There's a minor uptick in the early evening (7:00-8:00 PM), perhaps representing thoughts that percolated during the day and demanded expression before bed. But the pattern is unmistakable: the blog is written in the morning.

This aligns with conventional wisdom about creative work. The morning brain, rested and not yet depleted by decisions, handles complex thought better than the afternoon brain. Writers from Hemingway to Toni Morrison have described the morning as their productive window. The timestamps confirm that Cook follows the same pattern—whether consciously or not.

The timestamps appear to be in Central Time, consistent with Cook's location in Houston. A 7:00 AM post is genuinely early—not an artifact of timezone conversion.

The Yearly Shape

Monthly distribution reveals something unexpected: nothing.

Month	Posts	%
October	480	9.2%
November	478	9.1%
January	469	9.0%
September	448	8.6%
February	444	8.5%
August	438	8.4%
March	436	8.3%
July	410	7.8%
April	406	7.8%
December	417	8.0%
June	413	7.9%
May	394	7.5%

The range from highest (October, 9.2%) to lowest (May, 7.5%) is less than two percentage points. This is flat.

For comparison, consider what seasonal patterns typically look like. Academic blogs peak during semesters and crater in summer. News blogs spike during election seasons. Personal blogs often show holiday gaps. Most human activities have seasonal rhythms—productivity rises in autumn, slumps in late summer, crashes in December.

The blog shows none of this. October and January are essentially tied for the lead. May and July are essentially tied for the rear. The difference is statistical noise.

This flatness is remarkable. It suggests either extreme discipline (posting regardless of season) or complete integration (the blog doesn't compete with other activities that have seasonal patterns). Most likely it's both: a practice so embedded in daily routine that it neither competes with vacations nor depends on semesters.

Consistency Within Years

How steady is the monthly rhythm within each year?

One way to measure: the coefficient of variation (CV), which is the standard deviation divided by the mean. A low CV means even distribution across months; a high CV means lumpy distribution.

Year	Monthly CV
2010	0.14
2020	0.14
2008	0.19
2014	0.16
2017	0.20
2022	0.39
2015	0.38

Most years show CVs between 0.14 and 0.30—moderately even distribution. The blog doesn't have months that receive five times as many posts as others. The rhythm, while not metronomic, is steady.

The exceptions are interesting. 2022, despite being a strong year overall (362 posts), shows the highest variation (CV of 0.39). Some months got significantly more attention than others. 2015, a transitional year as the blog emerged from its 2014 dip, also shows high variation—the rhythm was unsettled.

But these are exceptions. The baseline pattern is consistency: month after month, year after year, the posts keep coming at roughly the same pace.

The Invisible Schedule

Cook wrote in 2024: "I've never had a schedule for this blog: I write when I feel like writing."

The data contradicts him.

He has no *{*calendar} in the sense of "I will publish every Tuesday and Thursday at 9:00 AM." But he has a *{*metabolism}.

The timestamps reveal a biological rhythm: write in the morning, publish most often on Tuesday, never stop for more than eleven days. The pattern is internal—habitual rather than calendared, felt rather than planned.

This is the difference between a schedule and a practice. A schedule is imposed from without; a practice grows from within. Cook doesn't experience himself as following a rule. But from the outside, the numbers trace one of the most consistent creative metabolisms documented on the internet.

Reading the Rhythm

Three patterns emerge from the temporal data.

{First: professional schedule.} The weekday dominance and morning concentration suggest that blogging is integrated into work time. It's not "I

write when I get home from my real job"; it's "writing is part of my real job."

{Second: internal rhythm.} The flat seasonal distribution suggests the blog runs on an internal clock, not an external calendar. Academic years don't shape it. Holidays don't disrupt it. The practice is autonomous.

{Third: evolved capacity.} The trajectory from 2008 to 2025 shows not a trade-off between quantity and quality but a growth in both. More posts *{*and} longer posts. The mature blog outproduces the young blog on both axes.

The data reveals a mind that shows up. Not every day, but most days. Not at the same minute, but in the same window. Not according to seasons or semesters or news cycles, but according to its own steady pulse.

Seven o'clock in the morning, usually Tuesday, regardless of month: that's when the writing happens.

The rhythm is steady. But what does that rhythm carry? The next chapters explore the content itself—the mathematical obsessions that fill those Tuesday morning hours.

Chapter 4

The Connectors

> "Think of the denominator of your fraction as something you have to buy. If you have enough budget to buy a three-digit denominator, then you're better off buying 355/113 rather than 314/100."
>
> "10 best rational approximations for pi" (2018)

Some numbers earn names. Some functions earn devotion.

The blog returns to certain mathematical objects again and again. Not because they are assigned by a curriculum, but because they are useful. These objects—pi, primes, the gamma function—are the fixed points of a mathematical imagination. They are the connectors that link one domain to another.

The Constants as Bridges

{Pi} dominates with 153 post titles.

The fascination spans seventeen years, from computational tricks ("Computing pi with bc") to surprising appearances ("The coupon collector problem and π").

Why pi? Not because of circles. Pi appears in the probability of a random walk returning to the origin. It appears in the distribution of prime numbers. It appears in the gamma function. Pi is a bridge: it connects geometry to probability to number theory. When Cook writes about pi, he is rarely writing about circumference; he is writing about connection.

{Primes} (103 post titles) connect the ancient to the urgent.

The blog traces the arc from Euclid ("multiply all primes together and add 1") to RSA encryption. Primes are the atoms of arithmetic, but they are also the foundation of digital security. Cook's treatment captures this dual nature: playful explorations of "Limerick primes" sit alongside serious discussions of primality testing for cryptography.

{Randomness} (95 post titles) connects the abstract to the concrete.

How do you generate points in Colorado? How do you simulate a parachute trial? Randomness is the tool for turning uncertainty into data. The blog's posts on Monte Carlo methods and quasi-random sequences are practical guides for using chaos to solve order.

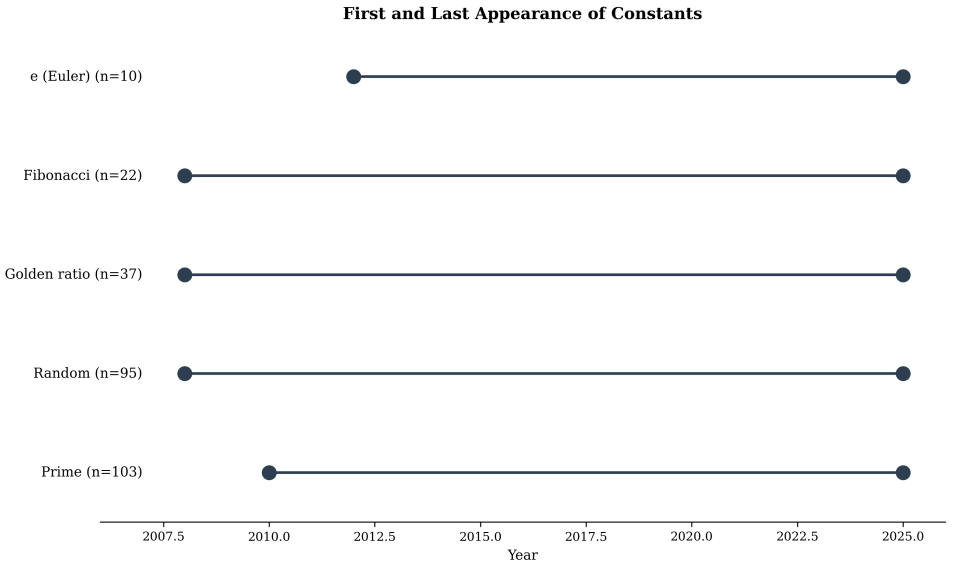


Figure 4.1: Mathematical constants: first and last appearances

The Alphabet of the Blog

The blog speaks in symbols. A frequency scan of the corpus shows how often key notation appears:

- {π} — 404 posts. - {$\sqrt{}$} — 333 posts. - {∞} — 136 posts. - {α} — 108 posts. - {β} — 83 posts. - {λ} — 81 posts. - {ϵ} — 70 posts. - {ζ} — 38 posts.

This is not just notation; it is a fingerprint. A blog about algebra would feature x and y. A blog about logic would feature $\forall$ and $\exists$. This blog features π, α, and ϵ—the symbols of analysis, probability, and approximation.

The Named Functions

A special function is just a function useful enough to deserve a name. Names emerge from use.

{The Gamma Function} (18 posts) is "the most important function not on a calculator."

It extends the factorial to non-integers. Because factorials appear in combinatorics, probability, and calculus, the gamma function appears in all

of them too. It is the blog's favorite tool for normalizing distributions and solving integrals. It bridges discrete mathematics (counting) and continuous mathematics (calculus).

{Fourier Transforms} (45 posts tagged "Fourier analysis") decompose the world into frequencies.

The 2023 post "Fixed points of the Fourier transform" asks: which functions equal their own transform? The normal distribution does. The hyperbolic secant does. The question reveals a deep structure connecting signal processing to probability theory.

{Bessel Functions} (12 posts) bridge physics and geometry. They describe heat in cylinders and vibrations on drumheads. A physicist needs them as often as a carpenter needs a hammer.

The tag "Special functions" marks 140 posts. These are the working tools of the trade.

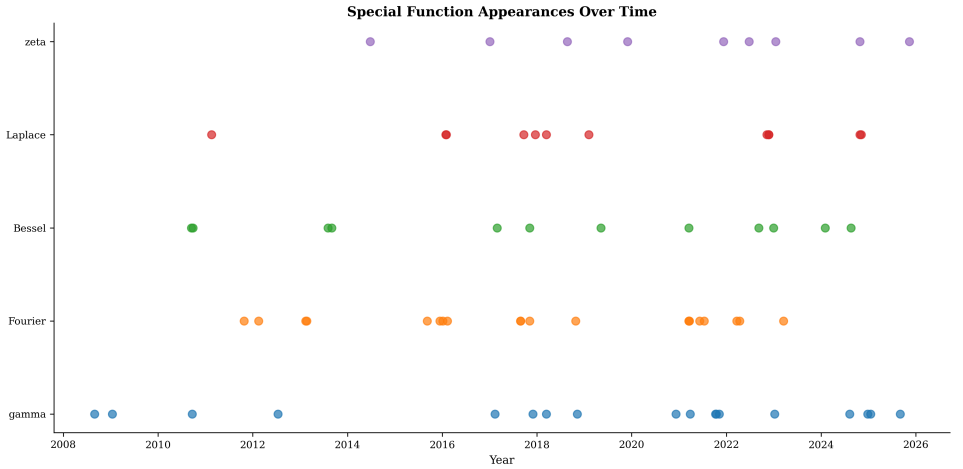


Figure 4.2: Special functions over time

The Web of Connection

The blog's obsession is not with the objects themselves, but with the web they form.

- Pi appears in the normal distribution (alongside *{*e}). - Primes underlie the security of random number generators. - Euler's formula links pi, *{*e}, and the imaginary unit. - The gamma function normalizes the beta distribution, which connects to Bayesian statistics.

The posts trace these lines. A 2025 post on pi approximations notes a formula hinting at Heegner numbers. The blog doesn't just share the formula; it notes the hint.

This is what a mathematical bestiary reveals: an ecosystem. The functions that recur are the ones that connect domains.

The blog's engagement with Ramanujan (10 posts) captures this. Ramanujan found simple formulas connecting unrelated constants. His pi approximation gives 18 correct decimal places using only roots and fractions. Cook quotes Hardy calling one result "most beautiful," then admits: "If I understood it, I might say it's beautiful, but for now I can only say it's mysterious."

Humility before difficulty, combined with persistent engagement—that is the stance.

Why These Objects?

A working mathematician returns to certain constants not for novelty, but for utility.

Pi solves problems about waves. Primes solve problems about encryption. The gamma function normalizes distributions. Each is a tool, sharpened over millennia.

The blog's treatment reflects this. Posts rarely introduce a constant from scratch. They encounter it in use—solving a problem, appearing in a formula, bridging two fields.

The reader learns pi not as "the ratio of circumference to diameter" but as "the thing that appears when you calculate the probability of a random walk returning to the origin."

That's what a mathematical imagination orbits: not the new, but the foundational. The numbers that keep appearing are the ones that keep being useful.

Chapter 5

The Translation Layer

> "I'd rather do math in a general language than do general programming in a mathematical language."
>
> "New introduction to SciPy" (2013)

Why do programmers read a mathematician's blog?

Because mathematics provides the abstractions, but code provides the leverage. The blog succeeds because it translates between the two.

The gamma function isn't just a concept; it's a line of Python. The symbol λ isn't just an eigenvalue; it's an anonymous function. The Fourier transform isn't just an integral; it's an algorithm. The blog sits at the translation layer, explaining the math to the programmers and the code to the mathematicians.

Python: The Lingua Franca

207 posts carry the "Python" tag. Python is the blog's native tongue.

The dominance isn't accidental. Python reads like pseudocode. A snippet explaining the gamma function is accessible to anyone with basic literacy:

```
from scipy.special import gamma
print(gamma(5))  # prints 24.0, which is 4!
```

The code is almost English. The mathematical content shines through.

The Python posts span the entire history, peaking in 2012 when Python was consolidating its role in science. By the 2020s, Python is assumed. It has become the background radiation of the blog's universe.

SciPy: The Toolbelt

80 posts tag "SciPy." This is the library that gives Python its mathematical teeth.

The blog documents the ecosystem not just by teaching it, but by testing it. "Bug in SciPy's erf function" (2010) documents a failure case for complex arguments. This is operational knowledge—the kind gained only by using the tools for real work.

The 2013 post "New introduction to SciPy" makes the thesis explicit: "The Python approach has its advantages—I'd rather do math in a general language than do general programming in a mathematical language."

This is the core argument. Python plus SciPy lets you do mathematics without abandoning general programming. You can read files, process data, and compute integrals in the same script. The cost is complexity; the benefit is power.

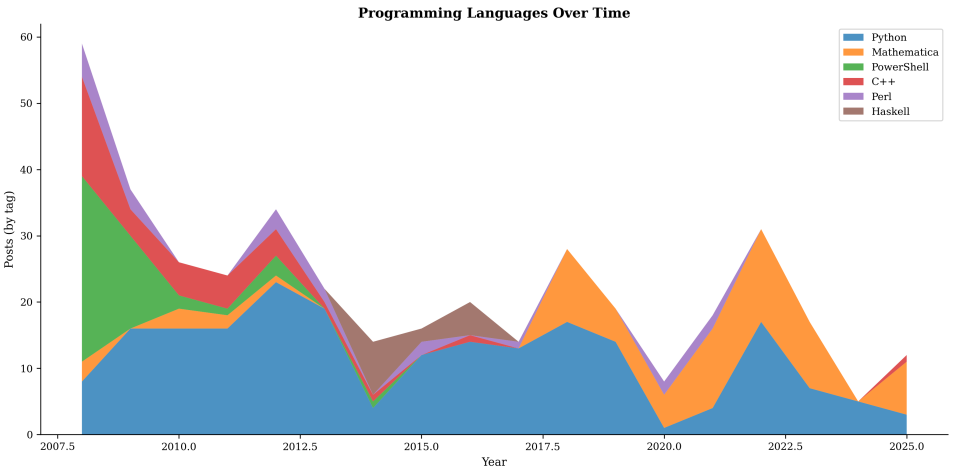


Figure 5.1: Programming languages over time

Mathematica: The Symbolic Engine

74 posts tag "Mathematica."

If Python is for building, Mathematica is for exploring. It handles symbolic manipulation—simplifying expressions, solving equations analytically.

The blog uses Mathematica to check its work. "DFT conventions: NumPy vs Mathematica" (2023) is a classic translation post: it compares how the two systems handle the Discrete Fourier Transform, documenting the subtle differences in convention that trip up practitioners.

Mathematica is the laboratory; Python is the factory. The blog moves between them.

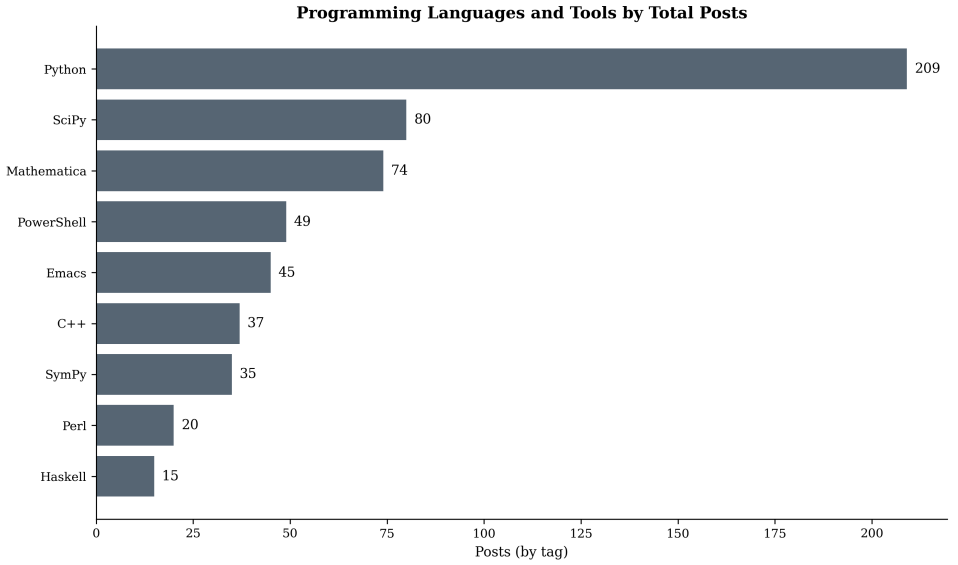


Figure 5.2: Programming languages by total posts

The Fossils: PowerShell and Perl

The data preserves extinct interests.

{PowerShell} (47 posts) dominates the early years (2008-2012), shows a lone straggler in 2014, and then disappears.

{Perl} (20 posts) fades similarly, with only sporadic appearances after 2015.

These fossils document the evolution of the field. In 2008, a working consultant needed PowerShell for Windows automation and Perl for text processing. By 2015, Python did both. The blog's topic list is a graveyard of tools that were replaced by better ones.

Code as Verification

Many posts include code not just to implement, but to verify.

"Let me check this numerically" is a recurring move. A post will derive a formula, then write a script to test it against a million random cases.

From "10 best rational approximations for pi" (2018):

> Here are the 10 best rational approximations to π, found via continued fractions.

The post includes the code. The code runs. The approximations work.

This is the empiricism of the computer age. Proofs are good; running code is better. The blog insists on both.

The Bridge

The blog's value lies in this bridge-building.

A programmer who needs the gamma function can look up the definition (useless) or find a library (opaque). The blog offers the third path: understand the concept *{*and*}* the implementation.

- Here's the math (gamma extends factorial). - Here's the code ({scipy.special.gamma}). - Here's the pitfall (singularities at negative integers).

This is the translation layer. It turns abstract theory into working machinery.

Chapter 6

Cryptography's Moment

> "Number theory was once the purest of pure mathematics—no applications, no use cases, beloved precisely because it was useless. Then it became the foundation of digital privacy."
>
> Adapted from G. H. Hardy's *A Mathematician's Apology*

Hardy died in 1947. Three decades later, RSA encryption turned prime numbers into the foundation of digital security. The purest mathematics became urgently practical.

The blog tracked this transformation in real time.

The Spike

Look at cryptography posts by year:

Year	Crypto Posts
2009-2016	~1/year
2017	5
2018	12
2019	**38**
2020-2024	~10/year
2025	47 (through November)

From 2009 to 2016: essentially nothing. Then 2017: the ramp begins. Then 2019: **{**thirty-eight posts}. In a single year.

Why 2019?

2019 was peak "techlash." Cambridge Analytica (2018). GDPR (2018). The world woke up to the reality of data surveillance.

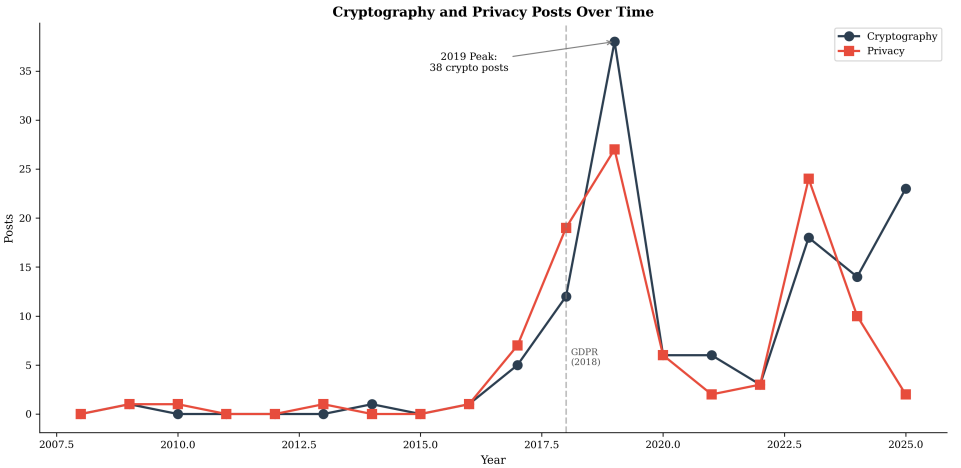


Figure 6.1: Cryptography and privacy posts over time

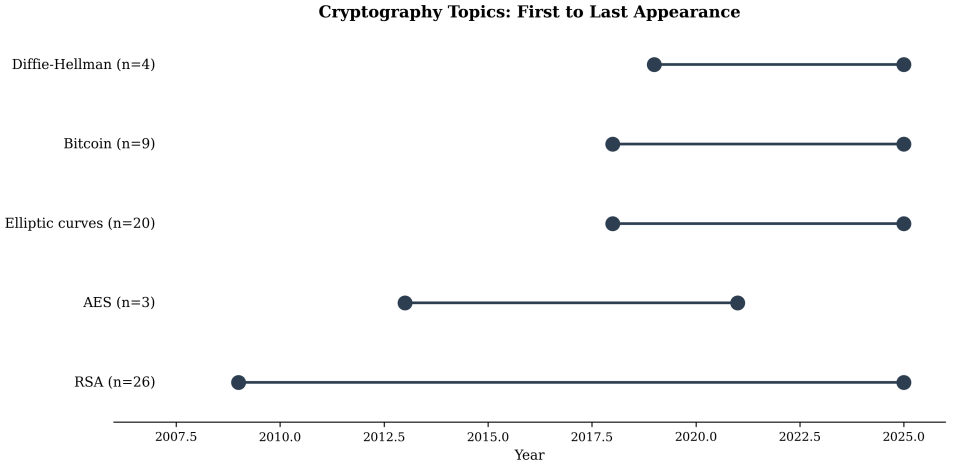


Figure 6.2: Cryptography topics: first to last appearance

Into this context, a mathematician who had been writing about number theory for a decade suddenly found his expertise urgent. Elliptic curves aren't just beautiful—they secure Bitcoin. Prime factorization isn't just a puzzle—it's why your passwords work.

The blog met the moment.

One post captures the tone: "Why RSA Works" (2019). It opens: "The security of RSA rests on the assumption that factoring large integers is hard." Then, in 600 words, it walks through the proof. No hand-wringing about privacy. Just: here is why the math works. The world was panicking; the blog offered clarity.

RSA: The Foundation

RSA encryption relies on the fact that multiplying primes is easy, but factoring the result is hard.

The blog explains RSA at every level: - **{**Theory:} Modular exponentiation. - **{**Practice:} Why *{e}* = 65537 is the standard exponent (it's 2^16 + 1, making it fast to compute). - **{**Failure:} How implementation flaws break the security.

The RSA posts span the blog's history, but they intensify when the world pays attention.

Elliptic Curves: The Modern Engine

25 posts tag "Elliptic curves."

The curve *{*$y^2 = x^3 + 7$*}* secures Bitcoin. Elliptic curves offer the same security as RSA with much shorter keys—essential for mobile devices and blockchains.

The blog's coverage clusters in two periods: - **{**2018-2019:} The cryptocurrency boom. - **{**2025:} Post-quantum concerns.

The 2018 post "Bitcoin key mechanism and elliptic curves" explains the math behind the money. It doesn't offer investment advice. It explains the group law on the curve.

Privacy: The Parallel Track

Privacy posts (27 in 2019) run parallel to the crypto posts.

But while crypto is about math, privacy is about law and logic. - **{**GDPR:} How to comply. - **{**HIPAA:} How to de-identify data. - **{**Anonymization:} Why removing names isn't enough.

This connects directly to Cook's consulting work. The "19th rule of HIPAA Safe Harbor" (2023) explains the catch-all clause that trips up compliance. This is where the mathematician meets the lawyer.

Pure and Applied

The blog resolves Hardy's dichotomy. Pure mathematics *{is}* applied mathematics waiting for its moment.

The 2018 post on RSA exponents illustrates this merger. It discusses properties of primes (pure) to explain why your bank connection is secure (applied). The distinction collapses.

When cryptography became urgent, the blog was ready. Not because Cook predicted the future, but because he had spent a decade explaining the foundations. When the spotlight turned to number theory, the stage was already set.

Chapter 7

The Edges of the Map

> "Why does music have a circle of fifths but no circle of thirds or circle of sixths?"
>
> "Circle of fifths and number theory" (2009)

The blog isn't only math.

At the center, yes—2,378 posts in the Math category, 45% of everything. But the edges tell another story. Music (85 posts). Typography (50). Clinical trials (55). Unicode (72). These are not mathematical categories by any standard definition. Still, they persist, year after year, for seventeen years.

What does a mathematician do when he's not doing math? The edges reveal the person.

Music: 85 Posts

The first music post appeared July 23, 2008: "Michael Brecker," a brief tribute to the jazz saxophonist. It's not mathematical at all—just an appreciation.

But by October 2009, music had become a mathematical topic: "Circle of fifths and number theory." The question is concrete: why can you cycle through all twelve notes of the chromatic scale by going up by fifths, but not by thirds or sixths? The answer involves modular arithmetic and the fact that 7 (the number of semitones in a fifth) is coprime to 12.

This is the pattern throughout. Music posts begin with a musical observation and arrive at mathematics:

- "Logarithms, music, and arsenic" (2008): Why musical intervals are measured in logarithms - "Mathematics and piano tuning" (2022): The compromises of equal temperament - "How many musical scales are there?" (2017): Counting with combinatorics - "The acoustics of kettledrums" (2016): Bessel functions in drumheads - "Dividing an octave into 14 pieces" (2022): Alternative temperaments

Nineteen posts carry both the Math and Music categories. These are the intersection points, where the two interests don't just coexist but merge:

- "Circle of fifths and roots of two" (2009) - "Perfect fifths, octaves, and an ergodic map" (2021) - "Overtones and Barbershop Quartets" (2025) - "Equipentatonic scale" (2025)

The mathematical content is real. Bessel functions describe vibrating circular membranes (drumheads). The overtone series is a Fourier decomposition. Equal temperament is a compromise between irrational and rational numbers—you can't have both perfect fifths and perfect octaves using the same tuning system.

Music appears because it can't not appear. The mathematics of sound, scale, and rhythm is too rich to ignore. A mathematician who plays or listens to music will eventually notice the structure.

Typography: 50 Posts

A mathematician who cares how symbols look.

Typography is about the visual representation of text: fonts, spacing, rendering, encoding. It's not traditionally part of mathematics, though mathematicians write formulas and care about their appearance.

The blog's typography posts cluster in the early years:

Period	Typography Posts
2008-2012	24
2013-2017	8
2018-2025	18

The early burst makes sense: Cook was establishing workflows, learning tools, sharing discoveries. "Accented letters in HTML, TeX, and MS Word" (2008) documents how to type ü and é across platforms—basic but constantly needed.

Later posts dig deeper:

- "Typesetting music with LilyPond" (2009): Where typography meets music - "Typesetting 'C#' in LaTeX" (2011): The sharp symbol has peculiar rendering - "Double-struck capital letters" (2023): The $\mathbb{N}$ and $\mathbb{R}$ of mathematical notation

The typography interest connects to precision. A mathematician who writes carefully about formulas will notice when the formula looks wrong. And a blogger who types thousands of posts will accumulate knowledge about character encoding, font rendering, and the quirks of different systems.

Clinical Trials: 55 Posts (2008-2018)

This category has a sharp edge: it ends.

Year	Clinical Trial Posts
2008	24
2009	6
2010	6
2011	2
2012	6
2013	1
2014	1
2015	5
2016	3
2018	1
2019-2025	0

Twenty-four posts in 2008—the launch year energy, but also clearly a professional focus. Then steady engagement through 2016. Then a final post in 2018. Then nothing.

The titles reveal practical work:

- "Cohort assignments in clinical trials" (2008) - "Stopping trials of ineffective drugs earlier" (2008) - "Bayesian adaptive clinical trials: promise and pitfalls" (2016)

This is applied statistics in medicine: how to design experiments that test drugs efficiently and ethically. The posts suggest someone working in pharmaceutical statistics, facing real design problems, writing about them.

The blog doesn't explain the break. There's no farewell post, no "I'm moving on from clinical work." The topic simply stops appearing.

This is a career transition visible in data. The shift from biostatistics to data privacy consulting left a clean break in the archive. One professional chapter closed; another opened.

Unicode: 72 Posts

An unexpected obsession.

Unicode is the standard for encoding text across languages and symbol systems. It specifies how computers represent everything from Latin letters to Chinese characters to mathematical symbols to emoji.

72 posts tag Unicode—more than Music or Typography. And the posts span the entire blog:

Period	Unicode Posts
2008-2012	11
2013-2017	7
2018-2022	37
2023-2025	17

The intensity increased over time. The recent era has more than three times as many Unicode posts as the early era.

Why? Several threads converge:

- Mathematical symbols live in Unicode. Writing about ∞ and $\sum$ and $\in$ means understanding their encodings. - International text processing requires Unicode. Anyone working with real-world data encounters encoding issues. - Unicode is peculiar. It has double-struck capital letters ($\mathbb{A}$, $\mathbb{B}$, $\mathbb{C}$), math alphabets, invisible characters, combining marks. A curious mind finds corners to explore.

Sample posts: - "Unicode arrows: math versus emoji" (2022) - "Unicode to LaTeX" (2013) - "Unicode spaces" (2020)

The Unicode interest sits at the intersection of computing and precision. You can't write mathematics on computers without understanding how computers represent mathematical symbols. You can't process text reliably without understanding encoding. Unicode is infrastructure—usually invisible, but occasionally demanding attention.

The Geography of Curiosity

30 posts carry the Geography category. These are mathematical explorations with spatial flavor:

- "Zip codes, geocodes, and Hilbert curves" (2022): Space-filling curves for spatial indexing - "State abbreviations as a random walk" (2018): Visualizing abbreviation patterns - "Generating random points in Colorado" (2025): Uniform sampling within arbitrary boundaries

Geography posts are sparse but persistent. They represent another edge where mathematics meets the physical world.

What the Edges Reveal

The edge categories share a pattern: they're not mathematics, but they involve mathematics.

Music involves frequencies, ratios, combinatorics. Typography involves encoding, rendering, visual precision. Clinical trials involve experimental design, Bayesian statistics, optimization. Unicode involves discrete structures, internationalization, edge cases.

The blog doesn't have a category for "pure hobbies"—no posts about sports, cooking, or travel as ends in themselves. The edges are edges of the mathematical map, not departures from it. They're places where mathematical thinking applies to non-mathematical domains.

This suggests something about the author's intellectual style: everything becomes a math problem. Hear a piece of music? Wonder about the frequency ratios. Encounter a character encoding issue? Investigate the standard. Design a clinical trial? Apply optimal stopping theory.

The edges aren't distractions from the center. They're extensions of it.

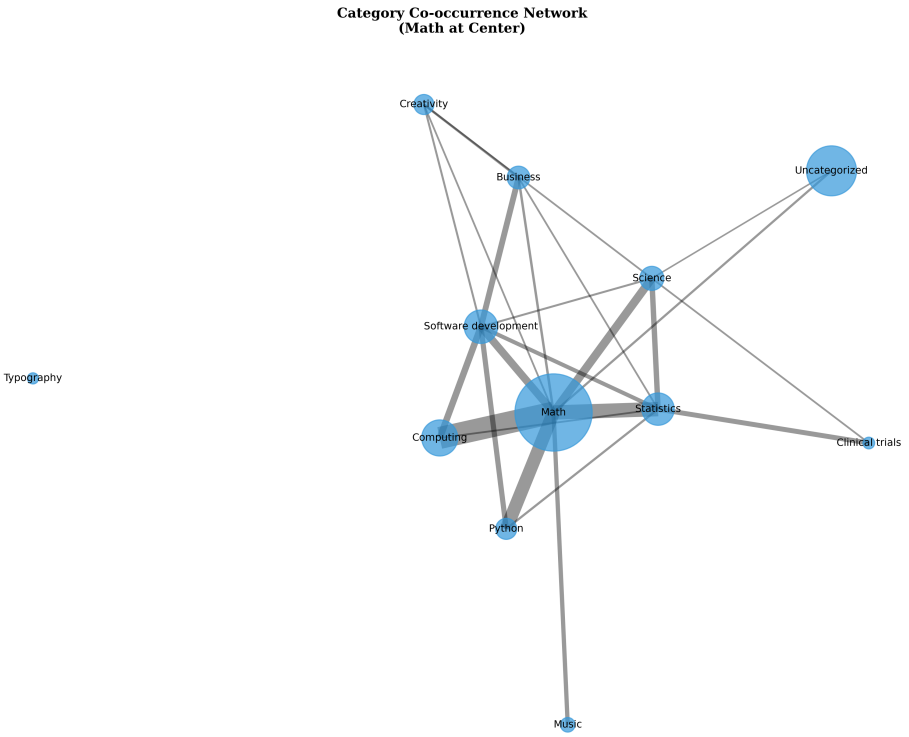


Figure 7.1: Category co-occurrence network

Humanization Through Data

The edges humanize the center.

A blog that was purely mathematical would be abstract, impersonal, a textbook in slow motion. The edges reveal a person: someone who listens to music, cares about fonts, worked in clinical statistics, and notices the details of character encoding.

The outline posed a question: "What does a mathematician do when he's not doing math?"

The answer from the data: he finds math in everything else. The edges of the map are not departures from mathematics but applications of it to domains mathematics doesn't usually claim. Music, typography, geography, encoding—all become mathematical when a mathematical mind engages with them.

The blog's edges are not escapes from the center. They're expeditions from it—and they always return.

The edges reveal a person behind the patterns. But another window opens onto that person: the conversations with others, documented across two decades of interviews.

Chapter 8

Input Mechanisms

> "When someone tells me a general theorem I say that I want an example that is both simple and significant."
>
> Sir Michael Atiyah, in conversation with John D. Cook (2013)

The blog is not a monologue. Twenty-two times, Cook handed the microphone to someone else.

These interviews, clustered between 2008 and 2013 with a few later podcast-style conversations, reveal the input side of the equation. Who does a mathematician listen to? What questions does he ask?

The list of subjects is a data point in itself.

The Builders

The interviewees share a trait: they build things.

{Dan Bricklin} built VisiCalc, the first spreadsheet. **{**Eric Floehr} built ForecastWatch, to measure weather accuracy. **{**Robert Ghrist} builds tools for applied topology.

Cook doesn't interview critics. He doesn't interview pundits. He interviews makers.

When he talks to Bricklin (2009), he asks: "What would your 30-second bio be without VisiCalc?" He wants to know about the person behind the artifact.

When he talks to Floehr (2011), he asks: "Did you start out intending this to be a business?" He wants to know how curiosity turns into infrastructure.

These questions mirror the blog's own trajectory.

The Boundary Crossers

The subjects refuse to stay in their lanes.

{Robert Ghrist} applies abstract topology to concrete sensor data. **{**Sacha Chua} blends programming with visual art. **{**Michael Atiyah} connects geometry to physics.

The blog values these intersections. The interviews act as validation: here are other people who think the way the blog thinks. Here are others who refuse to choose between "pure" and "applied."

The Atiyah Standard

The 2013 interview with Sir Michael Atiyah is the longest post in the archive (1,926 words). It serves as a manifesto.

Atiyah argues for "simple and significant" examples. "If there isn't a simple, interesting case, forget it."

This is the blog's editorial standard. Every post searches for that simple, significant example. The gamma function. The normal distribution. The circle of fifths.

Cook didn't just interview Atiyah; he adopted his worldview.

The Fading of the Format

After 2013, interviews become occasional rather than regular—an infrequent podcast, a news interview, a late conversation with a programmer. The intense interview phase fades, but the pattern never disappears entirely.

Why the shift? Perhaps the blog itself became the conversation partner. Perhaps the need to look outward was replaced by the momentum of the archive itself.

But the interviews remain as a record of formation. They show a mind seeking out its kin—finding the builders, the crossers, the people who make abstract things concrete.

They reveal that the "quiet institution" didn't grow in isolation. It grew by listening to the right people.

Chapter 9

The Evolution of a Blog

> "I started out posting a little more than once a day on average. I've slowed down a little... My posts are getting a little longer. They're still fairly short, but a little longer than they used to be."
>
> "3,000th blog post" (2019)

Seventeen years is long enough to see evolution.

The blog that launched in January 2008 is not the same blog that published its 5,233rd post in November 2025. Earlier chapters showed the patterns—the temporal rhythms, the topography of categories, the mathematical obsessions. This chapter asks a different question: what does the *{*change} mean?

From Notebook to Institution

The change isn't just about volume; it's about authority.

{The Notebook Phase (2008-2012):} The early blog reads like a consultant's scratchpad. "Here's a fix for a PowerShell bug." "Here's a link to an article." The voice is helpful but tentative. It's a professional sharing notes.

{The Clarification (2013-2017):} As the professional focus shifted, the voice tightened. The "tips and tricks" vanished. The mathematical explanations grew longer. The author stopped pointing at other people's work and started synthesizing his own.

{The Authority (2018-2025):} The modern blog doesn't just explain topics; it connects them. A post about the gamma function references a post about statistical mechanics which references a post about Python code. The voice is no longer just helpful; it is authoritative. The blog has become a reference work.

The arc is recognizable: experimentation, distillation, maturity.

What the Arc Reveals

This isn't just a blog getting older. It's a practice finding its form.

The early era shows someone writing about *{*everything}—useful observations from consulting work, mathematical curiosities, software tips, business insights. The middle era shows deliberate narrowing: what remains when professional obligations and passing interests are stripped away? Mathematics remains. The recent era shows what sustained focus produces: deeper posts, clearer explanations, the authority that comes from having explained the same territory for a decade.

The data confirms the depth. The median post length across the blog is 312 words, but the mean is 348. That 36-word gap reveals a long tail of deep-dive essays—the outliers that pull the average up. These are the posts where the blog stops being a notebook and starts being a textbook.

The 2014 dip—the quietest year—looks like a crisis from the outside. But it might have been clarification. The posts that year were fewer and shorter, and they sit on the hinge between the early notebook and the later deep dives—the calm before the length and depth of the later years arrive. Something was being decided about what the blog was *{*for}.

Compare a 2008 post to a 2023 post on the same topic. Consider the gamma function—a recurring obsession. In 2009, "The most important function not on a calculator" (256 words) offers a brief definition and a few applications. In 2023, "Maxwell-Boltzmann and Gamma" (635 words) derives the connection between the gamma function and statistical mechanics, includes Python code for verification, and links to three related posts. Same topic. Same author. Different voice—more confident, more substantive, more embedded in the archive's web of connections.

What Changed, What Didn't

{Changed:} - Post length doubled - Math percentage tripled - PowerShell, clinical trials, Perl disappeared - Cryptography emerged as a major theme - The professional consultant's voice faded

{Didn't change:} - Tuesday remained peak day - Morning remained peak time - Monthly distribution stayed flat - The core fascinations (pi, primes, special functions) persisted - The characteristic voice—concise, concrete, code-showing—remained recognizable

The consistencies matter as much as the changes. They suggest something persistent underneath the evolution: a mind that works a certain way, regardless of what it's working on.

The Lesson

The evolution shows: you don't need to know what you're building when you start.

The 2008 blog didn't know it would become a mathematical institution. It was a consultant's notebook with scattered interests and short observations. The purpose emerged through the work itself—through years of showing up, writing what seemed interesting, and gradually discovering what kept mattering.

This is how most sustained practices develop. The beginning is messy. The middle is clarification. The end (if there is an end) is confident focus.

The blog's seventeen-year arc is a case study in finding your subject by doing the work. Not by planning first and executing second, but by executing until the plan reveals itself.

For Creators

If there's a lesson that transfers, it's this:

{Start before you're ready.} The 2008 blog wasn't polished. It was exploratory, scattered, occasionally random. That was fine. The polish came later.

{Let interests compete.} The early era's breadth—software development, math, business, PowerShell—wasn't a mistake. It was an experiment. The middle era's narrowing happened because the competition produced a winner.

{Trust the process to clarify.} The 2014 dip wasn't failure; it was focus emerging. Sometimes fewer posts means better posts. Sometimes the quiet year is the year that matters most.

{Keep showing up.} The recent era's quality didn't come from talent or luck. It came from having explained the gamma function and the Fourier transform and elliptic curves so many times that the explanations became effortless.

The blog demonstrates what sustained practice produces: not just accumulated content, but accumulated capability. The 2025 voice is better than the 2008 voice because it has more practice behind it.

The Archive as Evidence

Unlike most creative development, this one is documented.

Usually, the messy early work disappears. Drafts get deleted. Early attempts get buried. The public sees only the polished result, with no trace of the stumbling that preceded it.

The blog preserves the stumbling. The short, scattered 2008 posts are still there. The PowerShell tutorials that no longer reflect current interests remain in the archive. The middle-era transition is visible in the changing ratios.

A reader who starts from the beginning witnesses seventeen years of development in compressed form. They see topics emerge and fade. They watch posts lengthen and deepen. They experience someone finding their subject.

This is rare evidence: what sustained intellectual work actually looks like, including the parts that don't fit the final shape.

Looking Forward

The data ends in November 2025, but the blog presumably continues.

What will the next decade show? Will new interests emerge, or will the mathematical focus deepen further? Will AI change how the blog operates—or become a topic it explains?

Whatever comes, it will build on seventeen years of established pattern. The blog knows what it is now. Future evolution will be variation on a settled theme, not the search for identity that characterized the early years.

For readers of this book, the question isn't what Cook will do next. It's what {you} will do.

The blog proves the thesis of this book's title: you can just do things. You can write about what interests you, publish without permission, show up day after day, and let the work find its form. You don't need a plan. You don't need a platform. You need time and consistency.

What will you build with your next seventeen years?

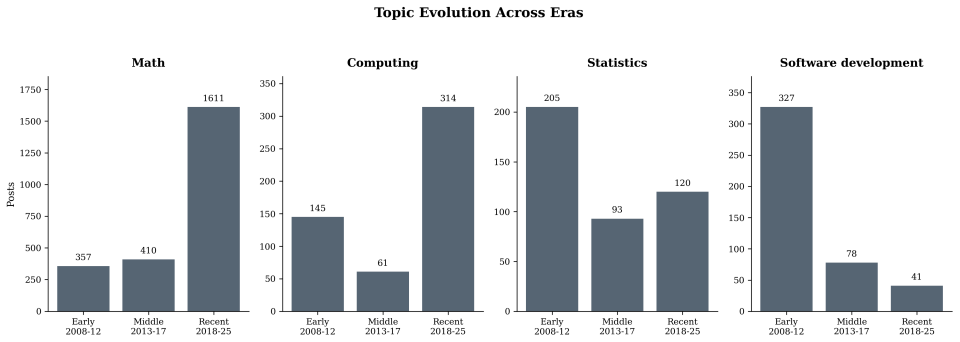


Figure 9.1: Topic evolution across eras

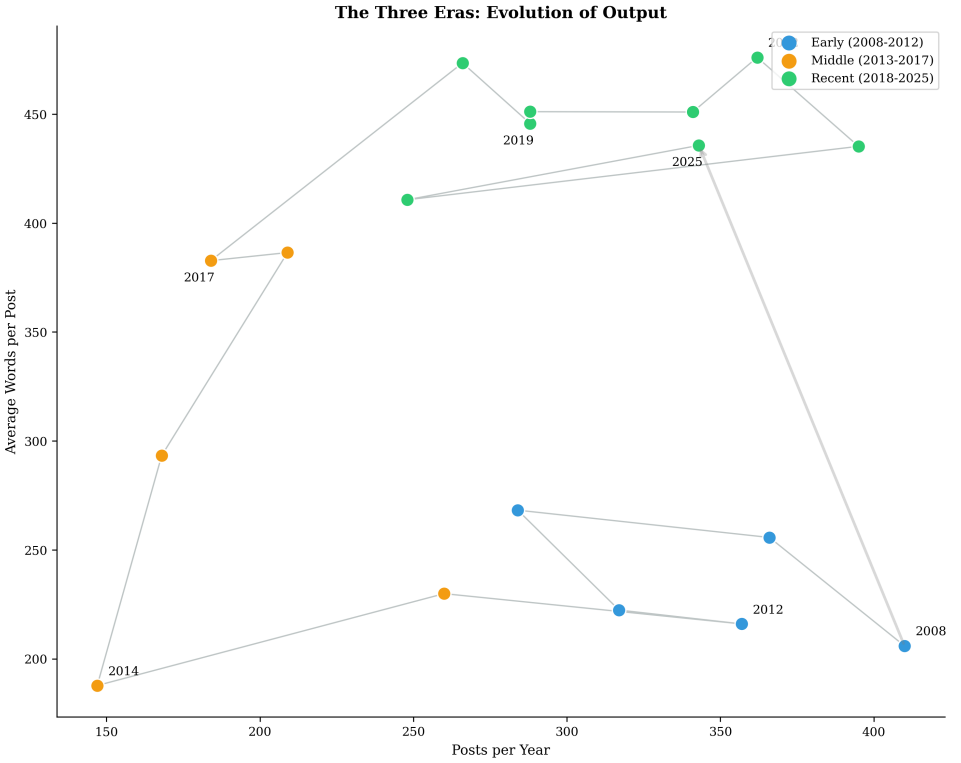


Figure 9.2: The three eras: evolution of output

Chapter 10

The Dark Matter

"The map is not the territory."

Alfred Korzybski

The numbers form the visible structure of the blog. But like dark matter in a galaxy, the invisible mass holds the system together.

5,233 posts. 26 categories. 1.8 million words. The numbers form a scaffold that supports everything else—the patterns, the eras, the evolutions, the spikes and fades. Without the counts, there's no story to tell.

But the most important things don't count.

The Influence Without Metrics

How many systems run code copied from this blog?

When a developer at 2 AM finds a post about the gamma function, understands finally what it does, and adapts the code for production—there's no metric for that. No download counter, no citation index, no way to trace the influence.

The blog has been a quiet presence in scientific computing for seventeen years. It explains special functions that standard documentation assumes you already know. It provides Python snippets for operations that should be obvious but aren't. It bridges the gap between mathematical definition and working code.

How many graduate students have that one post bookmarked? How many production systems contain a line adapted from an example here? How many hours of confusion has the blog prevented?

No number answers these questions. The influence is real but unmeasured—infrastructure without metrics.

The Signal in the Noise

The data captures everything, including the minimal.

"Music recommendations"—five words. The shortest post in the archive. It's a pointer, not an explanation.

232 posts fall under 50 words. 696 under 100. These are announcements, links, pointers—a sentence or two directing attention elsewhere. The data counts them equally with the deep dives. Each gets one entry in the post count.

But their function is different: not explanation but curation. The data treats every post as a unit of output, but some are units of connection.

The Roads Not Taken

79 tags appear exactly once in the entire archive.

Sage. Leadership. CSS. Decision theory. Neuroplasticity. Debugging. Etymology. Coq. Java. R. Agile. Humor.

Each represents a curiosity that didn't recur. Cook encountered Sage (the mathematical software), wrote one post, moved on. He wrote about leadership once—probably prompted by something specific, now forgotten. He mentioned Java exactly once in seventeen years.

The single-use tags form a long tail of momentary interests. They document what almost became a theme but didn't. They're the roads not taken, visible only because the tagging system preserved them.

What determines whether a curiosity becomes a recurring topic or stays a one-off? The data shows the outcomes but not the process. Why 25 posts about elliptic curves but only one about R? Why persistent engagement with Mathematica but a single mention of Sage?

The decisions happened in a mind, not a dataset.

The Missing Chorus

For years, blog posts accumulated comments. The comment sections were conversations, not monologues.

Eventually, the comments disappeared.

The archive now presents posts without their original discussion threads. Whatever insights readers contributed, whatever corrections they offered, whatever questions they asked—gone. The blog as it exists today is Cook's voice only, without the chorus that once responded.

The reasons—spam, moderation overhead—matter less than the result: a record that preserves the author's words but not the community's responses.

The Gravity of Influence

Which posts had the most impact?

The data offers proxies. Longest posts, perhaps. Most-tagged posts, maybe.

But these proxies miss the point. Impact happens in readers, not in metadata.

A 200-word post that finally clarifies the difference between the gamma function and the factorial might change how someone thinks forever. A 1,000-word post that covers familiar ground might change nothing. The word count can't distinguish.

The posts that mattered most are probably not the longest, not the most tagged, not the ones with "comprehensive" or "introduction" in the title. They're the ones that reached someone at the right moment—when they needed exactly that explanation, exactly that code snippet, exactly that connection between fields.

Those posts are invisible in the data. They're known only to the readers whose understanding they changed.

The Silent Infrastructure

The blog participates in a global ecosystem of technical knowledge. It is a resource consulted in moments of need, forgotten the next day, but crucial in that moment.

Some numbers do exist, even if they don't appear in the blog itself. Google Search Console reports readers in 244 countries. A single interpolation calculator page—one of more than eighty standalone tools Cook has built alongside the blog—has generated over 600,000 search clicks in the most recent 16-month Google Search window. That's one page, one tool, roughly a year and a half of traffic.

The blog has been submitted to Hacker News over 1,200 times, accumulating 25,000 upvotes and 10,000 comments. Four posts exceeded 500 points. These numbers capture a fraction of the readership—the fraction that uses Hacker News, that bothers to upvote, that comments. The actual reach is larger.

How many debugging sessions has this blog resolved? The number would be meaningful—it would measure real value, real hours saved, real problems solved. But it's unknowable. The search happens in private; the relief goes unrecorded.

A working engineer needs an answer: how do I compute this? A graduate student needs understanding: what is this? The blog serves both.

Academic citations occasionally capture this influence. Google Scholar reports nearly 6,000 citations for Cook's work—but most learning isn't cited. Most foundational understanding comes from sources that never

appear in bibliographies. A blog post that shaped how someone thinks about special functions might influence papers that never mention it.

The Territory vs. the Map

This book has drawn a map.

5,233 posts become patterns. The patterns suggest a person. The map is useful.

But the territory is larger.

The territory includes every reader who learned something. Every system running adapted code. Every student who understood finally. Every connection between fields that became visible because someone took the time to explain.

The territory includes the comment sections that no longer exist, the influences that were never counted, the changes in understanding that left no trace.

The territory includes the person behind the posts—the one who decided what to write, who chose examples and metaphors, who found simple ways to explain difficult things. The data shows the output; the mind that produced it remains opaque.

Why Count at All?

If the numbers miss so much, why count?

Because patterns suggest stories. Because stories suggest a person. Because even incomplete maps are useful.

The statistics in this book don't capture the blog's full influence. They can't measure impact, can't count understanding. But they reveal structure: what topics recur, how the blog evolved, what rhythms persisted, what changed.

That structure tells us something about seventeen years of intellectual work. Not everything—but something.

A mathematician starts a blog. He writes about what interests him. Some topics fade; others intensify. The posts get longer, deeper, more focused. A voice develops.

The numbers capture the shape of this evolution. They can't capture the meaning. But they can show that something happened here—something sustained, something coherent, something worth documenting.

The Blank Page

Every chapter in this book has included statistics.

This chapter has no visualizations. Deliberately.

Some things resist quantification. The influence of a good explanation can't be graphed. The value of a debugging session can't be tabulated. The long-term shaping of how someone thinks about mathematics—that's not a metric.

The data-heavy chapters earned the right to this data-free one. Having counted everything countable, the text can finally acknowledge what counting misses.

What Remains

The blog continues.

Readers will continue to find posts in moments of need. Students will continue to learn from explanations they'll never cite. Code will continue to flow from blog to production without acknowledgment.

The influence will remain unmeasured. The territory will remain larger than the map.

That's fine. Maps don't need to be complete to be useful. This one shows the shape of seventeen years of mathematical writing. It doesn't show everything; it shows enough.

The rest belongs to the readers who found what they needed, exactly when they needed it, and never thought to say thanks.

By the Numbers

This appendix is the canonical reference for all statistics in the book. Every number cited in the main text can be traced here. Data reflects the blog as of November 27, 2025; the archive continues to grow.

Overall Statistics

Metric	Value
Total posts	5,233
Total words	1,823,200
Date range	January 9, 2008 – November 27, 2025
Total days	6,531
Mean words per post	348
Median words per post	312
Longest post	1,926 words
Shortest post	5 words
Average posts per day	0.80
Average days between posts	1.25

Posts by Year

Year	Posts	Total Words	Mean Words	Median Words
2008	410	84,446	206	163
2009	366	93,597	256	194
2010	284	76,186	268	212
2011	317	70,508	222	185
2012	357	77,133	216	181
2013	260	59,789	230	172
2014	147	27,607	188	158
2015	168	49,277	293	238
2016	209	80,767	386	341
2017	184	70,425	383	339
2018	266	125,938	473	411
2019	288	128,358	446	414
2020	288	129,913	451	426
2021	341	153,769	451	433
2022	362	172,317	476	447
2023	395	171,900	435	424
2024	248	101,848	411	371
2025	343	149,422	436	413

Posts by Era

Era	Years	Posts	Mean Words	Total Words
Early	2008–2012	1,734	232	401,870
Middle	2013–2017	968	297	287,865
Recent	2018–2025	2,531	448	1,133,465

Categories

Category	Posts
Math	2,378
Uncategorized	996
Computing	520
Software development	446
Statistics	418
Business	206
Science	231
Creativity	161
Python	177
Music	85
PowerShell	47
Typography	50
Clinical trials	55
Graphics	43
Geography	30
Machine learning	24
AI	16
Artificial Intelligence	13

Note: Posts may have multiple categories. Compound categories (e.g., "Math;Python") are counted separately in the source data.

Top 100 Tags

Rank	Tag	Posts
1	Probability and Statistics	554
2	Math	498
3	Programming	342
4	Number theory	288
5	Python	207
6	Books	153
7	Special functions	140
8	Differential equations	134
9	Cryptography	128
10	Creativity	127
11	Productivity	123
12	Science	123
13	Geometry	110
14	Privacy	104
15	Bayesian	100
16	Complex analysis	92
17	SciPy	80
18	Numerical analysis	77
19	Mathematica	74
20	Unicode	72
21	Twitter	62
22	Integration	61
23	Music	61
24	Linear algebra	58
25	Quality	57
26	Clinical trials	56
27	LaTeX	55
28	Education	55
29	Networks	54
30	Category theory	52
31	History	51
32	Signal processing	49
33	PowerShell	49
34	Regular expressions	46
35	Fourier analysis	45
36	Emacs	45
37	Group theory	44
38	Orbital mechanics	42
39	Command line	42
40	Coding theory	41
41	Mental math	40
42	RNG	40
43	Combinatorics	40
44	Rstats	40
45	Quotes	40
46	C++	37
47	SymPy	35
48	Economics	35
49	Logic	35
50	Cryptocurrency	34
51	Acoustics	34

Posts by Day of Week

Day	Posts	Percentage
Monday	887	17.0%
Tuesday	1,003	19.2%
Wednesday	938	17.9%
Thursday	823	15.7%
Friday	561	10.7%
Saturday	555	10.6%
Sunday	466	8.9%

Posts by Hour of Day

Hour	Posts	Percentage
00:00	12	0.2%
01:00	13	0.2%
02:00	11	0.2%
03:00	25	0.5%
04:00	73	1.4%
05:00	179	3.4%
06:00	501	9.6%
07:00	683	13.1%
08:00	580	11.1%
09:00	503	9.6%
10:00	323	6.2%
11:00	315	6.0%
12:00	245	4.7%
13:00	234	4.5%
14:00	218	4.2%
15:00	194	3.7%
16:00	176	3.4%
17:00	159	3.0%
18:00	139	2.7%
19:00	202	3.9%
20:00	221	4.2%
21:00	141	2.7%
22:00	67	1.3%
23:00	19	0.4%

Peak hour: 07:00 (683 posts, 13.1%)

Mathematician Mentions in Titles

Mathematician	Posts
Fourier	20
Gauss	16
Laplace	12
Bessel	12
Newton	11
Euler	10
Ramanujan	10
Fermat	8
Riemann	7
Knuth	6
Cauchy	5
Hilbert	5

Special Functions in Titles

Function	Posts
Gamma	18
Bessel	12
Zeta	9
Hypergeometric	9
Legendre	4

Programming Languages

Language	Tag Count
Python	207
SciPy	80
Mathematica	74
PowerShell	49
C++	37
SymPy	35
Perl	20
Haskell	15
Lisp	10
R (Rstats)	40

Extremes

Longest Posts

Rank	Title	Date
1	Interview with Sir Michael Atiyah	2013-09-24
2	Applied topology and Dante: an interview with Robert Ghrist	2010-09-13
3	Hallucinations of AI Science Models	2024-03-26
4	Field of order 9	2022-09-21
5	Writing down an unwritten language	2014-02-24
6	Interview with author Cliff Pickover	2009-08-30
7	Anatomy of a posit number	2018-04-11
8	Solving for Möbius transformation coefficients	2021-03-28
9	Two-sample t-test and robustness	2018-05-11
10	Probability that a number is prime	2010-10-06

Shortest Posts

Rank	Title	Words
1	Music recommendations	5
2	Test post	7
3	Synchronizing metronomes	7
4	New prime number record	9
5	PowerShell A-Z series completed	9
6	Introduction to computer virtualization	10
7	Simple interpolation	11
8	Technology hype curve	12
9	Getting started with SciPy (Scientific Python)	12
10	51st Carnival of Mathematics posted	13

Key Dates

Event	Date
First post	January 9, 2008
Post #1,000	~2010
Post #2,500	~2016
Post #5,000	~2025
Last post in dataset	November 27, 2025

Single-Use Tags

79 tags appear exactly once in the entire archive. A sample:

Sage, Leadership, CSS, Decision theory, Neuroplasticity, Debugging, Etymology, Coq, Java, R, Agile, Humor, AMD, CUDA, GPU, Clojure, Convergence, Z3, Vista, GPT-5, Automated reasoning, Integral equations, NP-completeness, Numerica, Ring theory...

Each represents a curiosity that was explored once and not repeated.

Data Notes

- Word counts are approximate, extracted via text processing - Categories and tags are as assigned by the author - Timestamps are in Central Time (US) - Some HTML entities appear in titles (e.g., ’ for apostrophe) - Data extraction completed November 2025

A Reading List

Fifty posts worth reading, curated from 5,233.

This list draws from the longest posts, the most cross-disciplinary, the signature topics, and the interviews. It is not comprehensive—nothing could be—but it provides entry points across the blog's seventeen-year range.

All posts are available at johndcook.com/blog.

Start Here: 10 Essential Posts

New to the blog? Start with these ten posts, chosen to represent the range and character of seventeen years of writing.

Date	Title
2013-09-24	Interview with Sir Michael Atiyah
2010-10-06	Probability that a number is prime
2009-10-02	Circle of fifths and number theory
2018-09-04	Pi primes
2023-03-17	Fixed points of the Fourier transform
2025-11-16	Elliptic curve pairings in cryptography
2016-02-20	The empty middle: why no one is average
2020-04-19	LaTeX command frequencies
2011-04-12	Evaluating weather forecast accuracy: an interview with Eric Floehr
2008-01-09	Moore's law and software bloat

The Interviews

The blog's conversations with remarkable people. These are often the longest posts in the archive.

Date	Title	W
2013-09-24	Interview with Sir Michael Atiyah	1
2010-09-13	Applied topology and Dante: an interview with Robert Ghrist	1
2009-08-30	Interview with author Cliff Pickover	1
2013-03-19	Interview with Sacha Chua	1
2011-04-12	Evaluating weather forecast accuracy: an interview with Eric Floehr	1
2012-07-30	Interview with Michael Hammel, author of The Artist's Guide to GIMP	1
2016-06-10	Interview with Chris Toomey of Upcase	1
2009-06-22	Dan Bricklin interview	1
2009-07-29	Carl Franklin interview	8
2010-03-08	Interview with Clojure author	4

The Longest Posts

Deep dives that grew beyond the typical length. Two of the top ten are interviews.

Date	Title	Words
2024-03-26	Hallucinations of AI Science Models	1,710
2022-09-21	Field of order 9	1,695
2014-02-24	Writing down an unwritten language	1,601
2018-04-11	Anatomy of a posit number	1,575
2021-03-28	Solving for Möbius transformation coefficients	1,456
2018-05-11	Two-sample t-test and robustness	1,429
2010-10-06	Probability that a number is prime	1,390

Pi: The Constant

The most-mentioned number in the archive, explored from multiple angles.

Date	Title	Words
2018-09-04	Pi primes	548
2021-02-12	Finding pi in pi with Perl	488
2021-10-07	Gamma and the Pi function	451
2017-10-23	Finding numbers in pi	891
2011-03-14	Algorithm used for world record pi calculations	406

Primes: The Building Blocks

The integers that divide only by themselves and one—explored for seventeen years.

Date	Title	Words
2010-10-06	Probability that a number is prime	1,390
2025-08-11	Memorable Primes	434
2024-02-10	Factoring pseudoprimes	790
2023-09-26	Analogy between prime numbers and simple groups	576
2024-12-31	Details of generating primes for cryptography	410

The Gamma Function

The most important function not on a calculator.

Date	Title	Words
2025-01-16	Dimensional analysis for gamma function values	621
2023-01-07	Maxwell-Boltzmann and Gamma	635
2021-11-06	Gamma distribution tail probability approximation	686
2021-03-26	Simple approximation for Gamma function	449

Fourier Analysis

Decomposing signals into frequencies—the workhorse of modern engineering.

Date	Title	Words
2023-03-17	Fixed points of the Fourier transform	460
2022-04-13	Hilbert transform and Fourier series	645
2022-03-20	Reverse engineering Fourier conventions	780
2021-06-07	Fourier, Gauss, and Heisenberg	421

Cryptography

The 2019 spike and beyond—number theory made urgent.

Date	Title	Words
2025-11-16	Elliptic curve pairings in cryptography	671
2025-11-17	Three-party Diffie-Hellman in one shot	513
2025-09-30	Time needed to factor large integers	487
2025-10-06	A quiet change to RSA	449
2025-11-18	RSA as a pairing	423

Statistics

Probability and inference—the second-largest domain.

Date	Title	Words
2018-05-11	Two-sample t-test and robustness	1,429
2025-04-28	Variance of variances. All is variance.	614
2016-02-20	The empty middle: why no one is average	547
2009-11-24	A beta-like distribution	553

Music

Where mathematics meets sound.

Date	Title	Words
2009-10-02	Circle of fifths and number theory	706
2009-03-15	Typesetting music with LilyPond	612

Computing and Programming

The bridge between mathematics and machines.

Date	Title	Words
2020-04-19	LaTeX command frequencies	514
2021-10-02	Tables and interpolation	433
2009-01-23	Probability distributions and object oriented programming	947

Cross-Disciplinary Posts

Posts spanning multiple categories—the connections between fields.

Date	Title	Categories
2025-08-27	Why are CUDA kernels hard to optimize?	AI, Computing, HPC,
2025-07-09	Experiences with Nvidia	AI, Computing, HPC,
2025-06-05	Scientific papers: innovation … or imitation?	AI, Creativity, Produc
2008-11-17	Fast way to test whether a number is a square	Computing, Math, Sc

Science and the Physical World

Where mathematics touches reality.

Date	Title	Words
2017-04-09	Mercury and the bandwagon effect	476
2011-09-22	Big data and humility	353
2010-02-19	Popular research areas produce more false results	313

How to Use This List

This list represents roughly 1% of the archive. It favors longer posts, which tend to be more developed, and posts that span categories, which tend to be more unusual.

For any specific topic—quaternions, continued fractions, Unicode, orbital mechanics—search the blog directly. The archive is searchable at johndcook.com/blog, and seventeen years of posts mean that most mathematical and computational topics have been touched at least once.

The interviews are particularly recommended as entry points. They reveal both the subjects' thinking and the interviewer's interests—what he chose to ask, where he pushed for detail, what connections he noticed.

For readers who want to follow the blog's evolution, reading chronologically from 2008 is illuminating. The early posts are shorter and more scattered; the recent posts are longer and more focused. Watching the transformation unfold gives a sense of intellectual development that no single post can provide.

A Note on Selection

Any list of fifty posts from 5,233 involves judgment calls. These selections favor:

- Length (longer posts generally contain more developed arguments) - Cross-disciplinarity (posts spanning categories are rarer and often more interesting) - Signature topics (pi, primes, gamma, Fourier—the recurring obsessions) - Interviews (the only posts featuring other voices)

A different curator would choose different posts. The archive is deep enough to support many valid reading lists. This one provides starting points; the exploration is up to the reader.

How This Book Was Made

This is a book about a mathematician's blog, made by counting things. It seems only fair to explain how the counting was done.

The Data

Everything derives from publicly available data. The blog at johndcook.com runs on WordPress, which means every post has structured metadata: title, date, categories, tags, word count. This metadata was extracted, cleaned, and stored in a few primary files—5,233 rows mapping dates to titles to word counts to tags.

No private data was used. The blog is public; this work simply organized what was already there.

The Tools

{Python 3.12} for all data processing—the same language Cook has written about extensively.

{pandas} for data manipulation. **{**matplotlib} and **{**seaborn} for visualization. Every chart in this book was generated programmatically; the code is reproducible.

{Pandoc} for document compilation. Markdown to PDF, with LaTeX for typesetting.

The choice to use Python was partly practical and partly fitting: analyzing a blog with tools the blogger helped people understand.

The AI Assistance

This book was written with substantial assistance from Claude, an AI made by Anthropic. Not as a ghostwriter—every word was reviewed, edited, and approved by humans—but as a collaborator for exploration, drafting, and iteration.

The repository root contains a {CLAUDE.md} file with instructions that persist across sessions: coding conventions, file organization, project context. This is "spec-driven development"—a specification document that compounds across sessions, each one starting with full context rather than a blank slate.

The rhythm was collaborative. The human provides direction and judgment. The AI provides speed and tirelessness. Neither could produce this book alone.

The Irony

There's an irony worth naming.

This is a book celebrating seventeen years of one person's writing. Cook wrote 1.8 million words by hand, one post at a time. No shortcuts. Pure human persistence.

The irony: a book on human persistence, built with AI speed. Visualizations that would have taken hours appear in seconds. Drafts that would have taken days arrive in minutes.

Is this cheating? Is it homage?

Cook himself has written about AI—about automation and its limits, about the boundary between human and machine intelligence. He's not a Luddite. He's curious about tools.

What the AI Can't Do

Certain things remained stubbornly human:

{Judgment.} The AI can generate twelve visualization types. It can't tell you which one serves the chapter's argument. That requires understanding what the book is *{*for}.

{Taste.} The first visual sampler was four pages and felt like a tech demo. A human said "this won't work." The AI couldn't have made that call.

{Structure.} The original outline had eighteen chapters. It was a data dump, not a book. A human recognized this.

{The prologue.} The AI wrote drafts. But the decision about what the prologue should *{*do}—establish significance, justify the book's existence, set the tone—that was human.

This book is human-directed and AI-assisted. The shape, the purpose, the reason it exists: those are human.

A Different Kind of Methodical

This project is the mirror image of the work it describes.

The blog was built slowly, one small post at a time, with no master plan beyond showing up and explaining something clearly. This book came together quickly once the data and tools were in place—a compressed burst made possible only because those seventeen years of methodical writing existed to analyze.

The contrast says something about human-generated content in an era of automation. As models improve, it becomes easier to synthesize and visualize what already exists. What does not get easier is producing the raw material worth summarizing: the careful explanations, original examples, and durable insights that give these tools anything meaningful to work with.

Cook's archive is a reminder that even in an era of AI, the slow accumulation of clear human work is what matters most.

Closing Thoughts

Cook has written about the tools we use and how they shape our thinking. Python as a thinking tool. LaTeX as a writing environment. The way software encodes assumptions.

This book was made with tools he would recognize. Python, pandas, matplotlib—the same stack he's explained in dozens of posts. If there's a fitting tribute to a technical blogger, it might be this: the tools he taught others to use were turned back on his own work.

Whether that's homage or irony is left to the reader.

And whether the method transfers—whether you, too, can just do things, can build something useful by showing up day after day—that's left to you.

This epilogue was drafted with AI assistance and edited by humans. The full editorial process—including assessments from three AI editors (Claude, Codex, and Grok) and two revision passes—is documented in the 'editorial/' folder.